Example 1: Let V = 10 V,
I=2 A, find the value of P.

$$P = V \times I = 10 \times 2 = 20 \; watt$$

Example 2: Let I = 2A, R=15Ω, find the value of P.

$$P = I^2 R = 2 \times 2 \times 15 = 60 \; watt$$

Example 3: Let V = 6 V, R=12Ω, find the value of P.

$$P = \frac{V^2}{R} = \frac{6 \times 6}{12} = 3 \; watt$$

1.4.2 Electromotive Force (E.M.F)

The force exerted by the density of the electrons in a point is referred as Electromotive Force (E.M.F). Also, it is the voltage developed by any source of electrical energy such as battery or dynamo. Its unit is Volt.

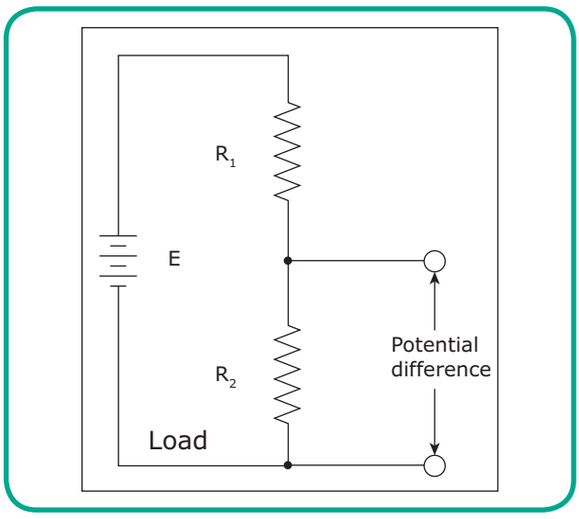


Figure 1.9 Potential difference across the load

1.4.3 Potential Difference

It is the voltage across any two terminals of a load in a circuit as shown in Figure 1.9. It is measured in Volt.

1.4.4 Electrical Load

If electrical energy is supplied to an electrical appliance, it starts operating using that energy. For example, an electrical fan supplied with electrical energy starts rotating and gives air circulation. In this phenomenon, fan is a load, consuming the electrical energy. Similarly, electric lamp offers light using electrical energy. Here, lamp acts as a load by consuming the electrical energy. From this we come to know that

An object that consumes electrical energy is called electrical load. e.g., electrical lamp, fan, mixie, radio, television, etc.

1.4.5 Electrical Circuit

If two terminals of an electrical supply is connected to a load, which enable to pass current through the electrical devices is called electrical circuit as shown in Figure 1.10.

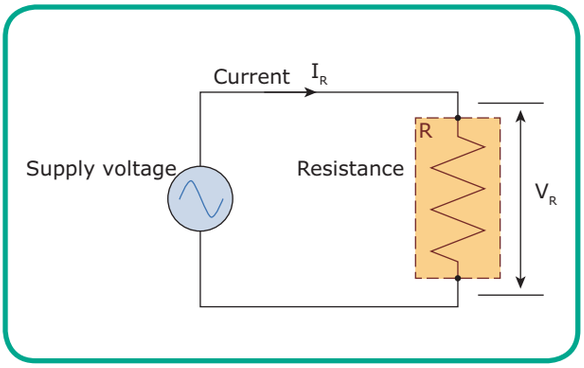


Figure 1.10 Electrical circuit

1.4.6 Symbols

One cannot draw the actual appearance of electrical connections and loads. Hence, simplified symbols are allotted to electrical connections, loads, appliances and devices. For example, few symbols are given in Figure 1.11.

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